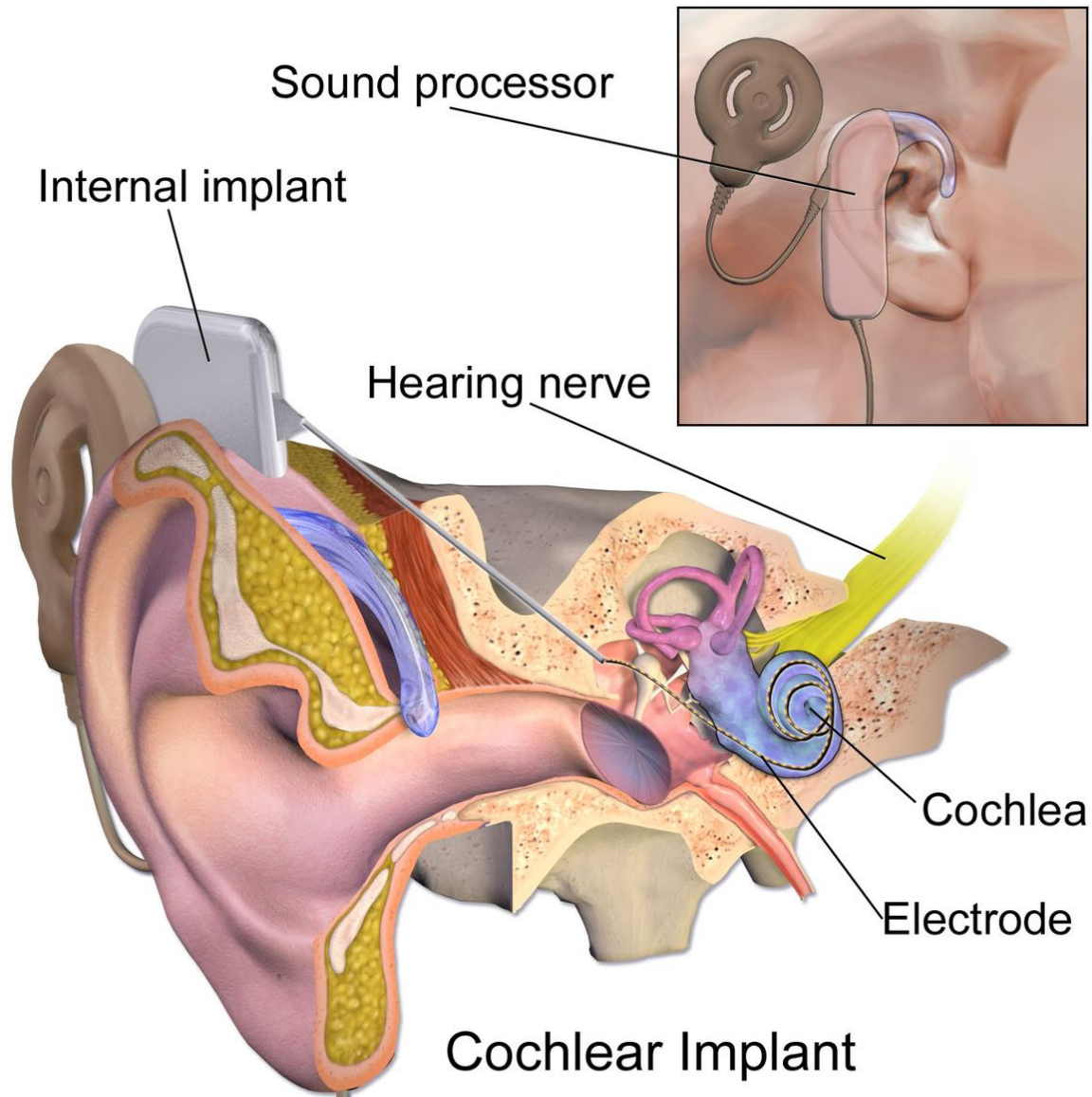




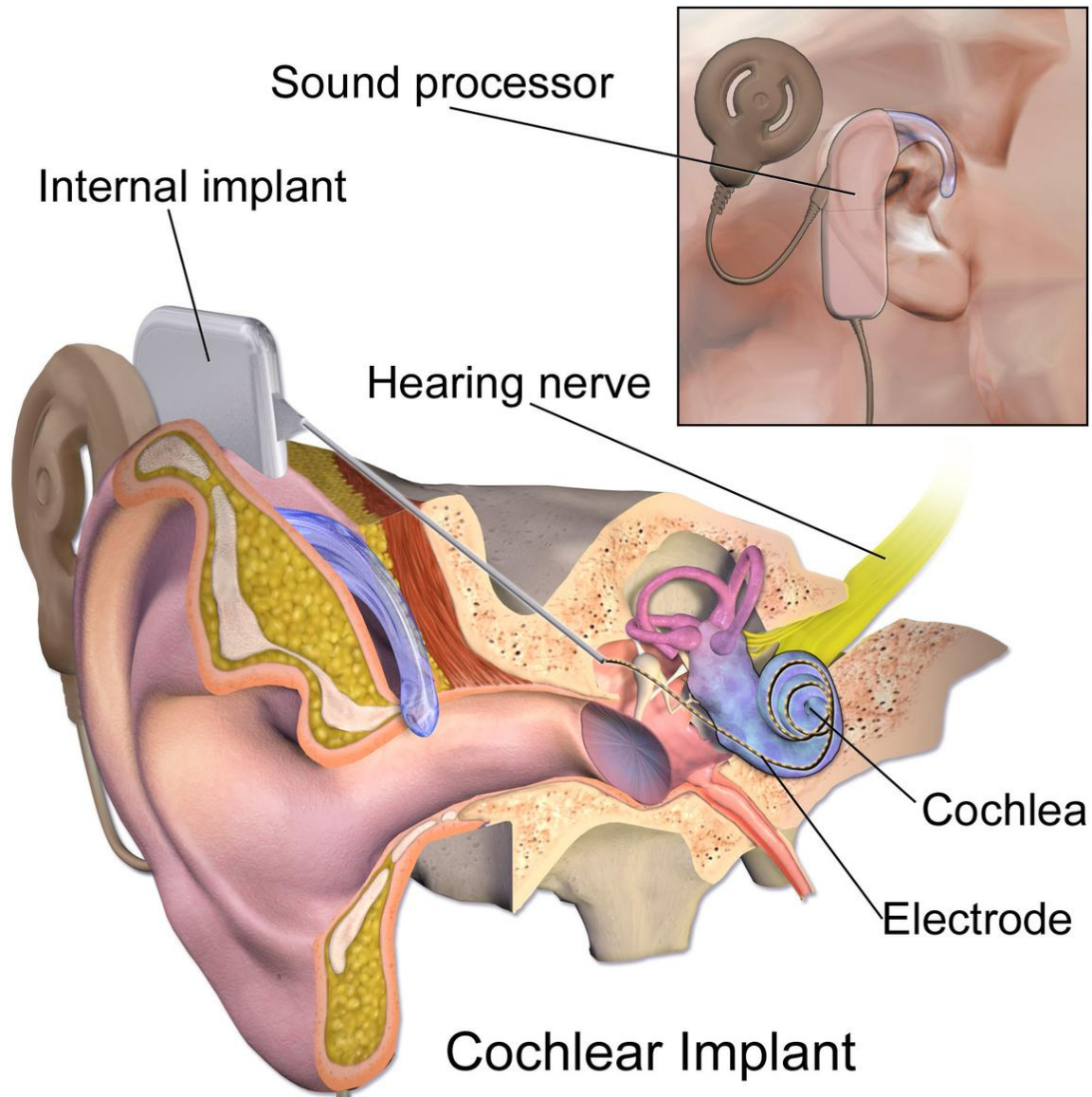
Neural Modelling of complex pulse train stimulation to improve cochlear implant coding strategies

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Laboratory for Experimental Audiology, ORL-USZ

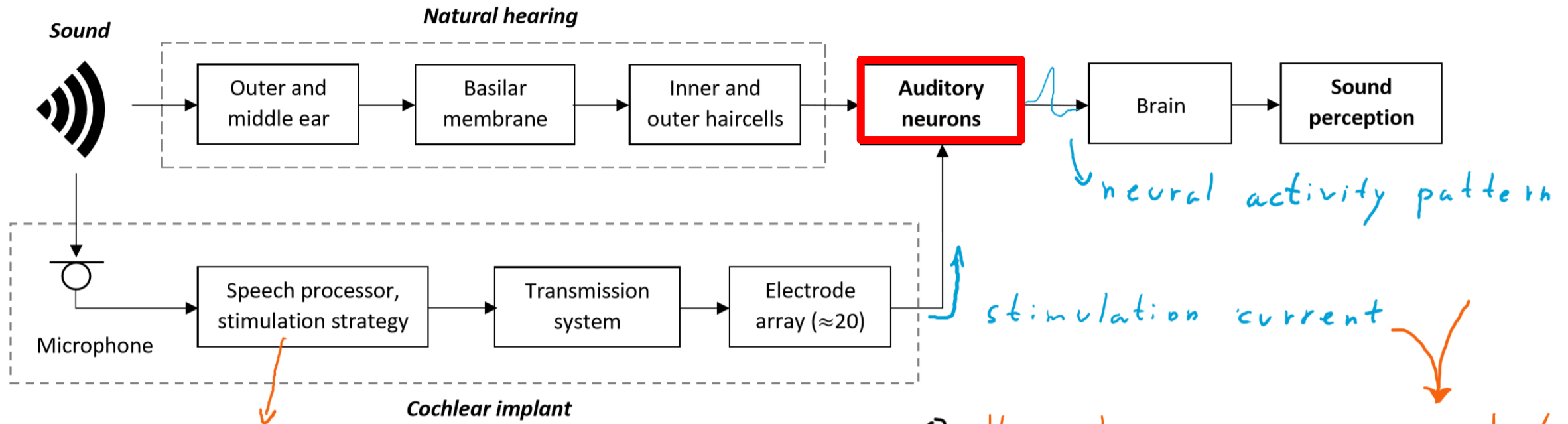


Hearing loss and cochlear implants

- Hearing is crucial in work, school, daily life... sadly, many suffer from hearing loss:
 - *Disabling hearing loss (>40 dB): ~500 milion*
 - *Severe hearing loss (>80 dB): ~40 milion*
 - *Cochlear implant patients: ~400.000*
- **Cochlear implant (CI):** restores hearing by electrically stimulating auditory neurons
- Hearing is restored, but many issues remain:
 - hearing in noise, music, tonal languages...
- **Reasons behind these issues?**



Coding strategies neglect (nonlinear) neural dynamics → reduced hearing quality
Idea: Develop a *simple model* of neural dynamics *usable* in coding strategies



Q: Which stimulation current pattern to use for a given sound?
→ coding strategy!

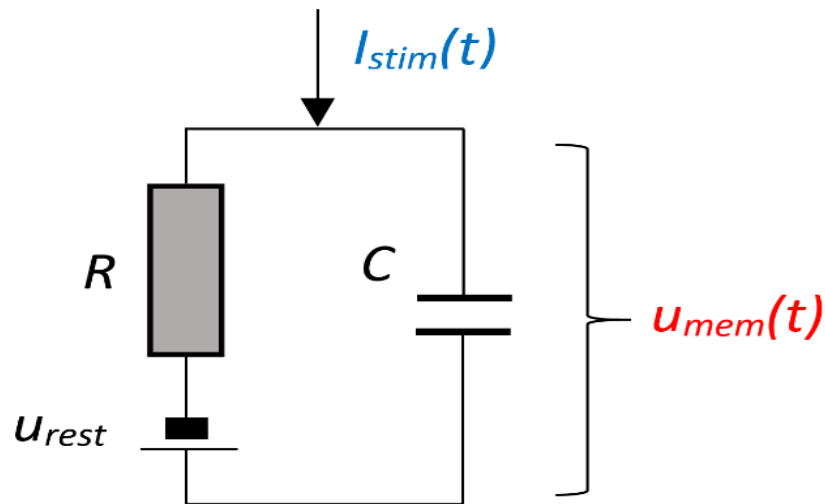
Q: How do neurons respond to the applied stimulation current?

→ Develop a neuronal model!

Model: Integrate & Fire model with dynamic threshold

Single neuron

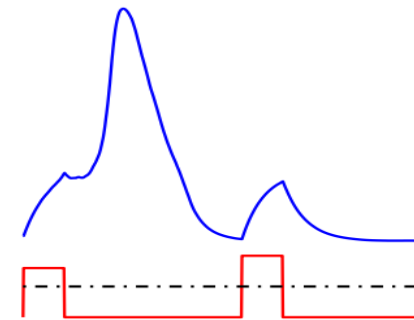
- Charge-discharge behavior (RC circuit)
- **Spike when voltage is above threshold**
- Noisy threshold



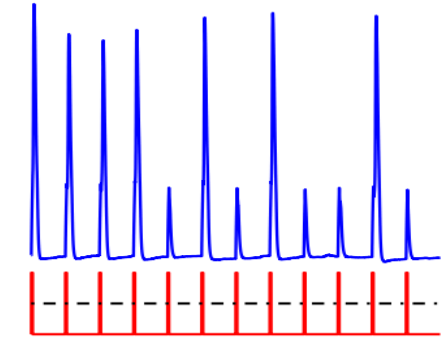
Neuronal population = single neuron model repeated + **electrical stimulus spread** model

Nonlinear neural dynamics included by **modifying the voltage threshold**

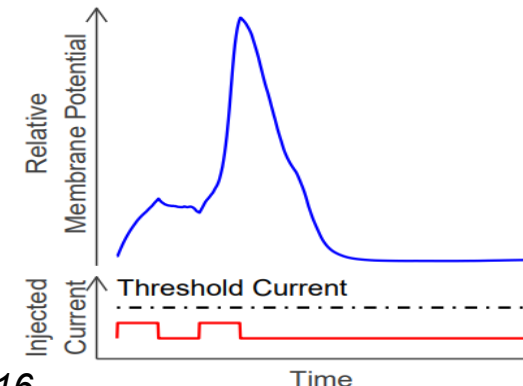
A Refractoriness



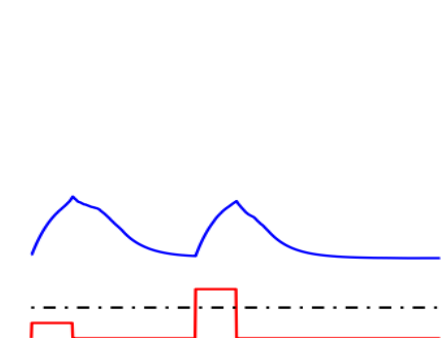
B Spike-Rate Adaptation (Spike-Dependent)



C Facilitation (Temporal Summation)



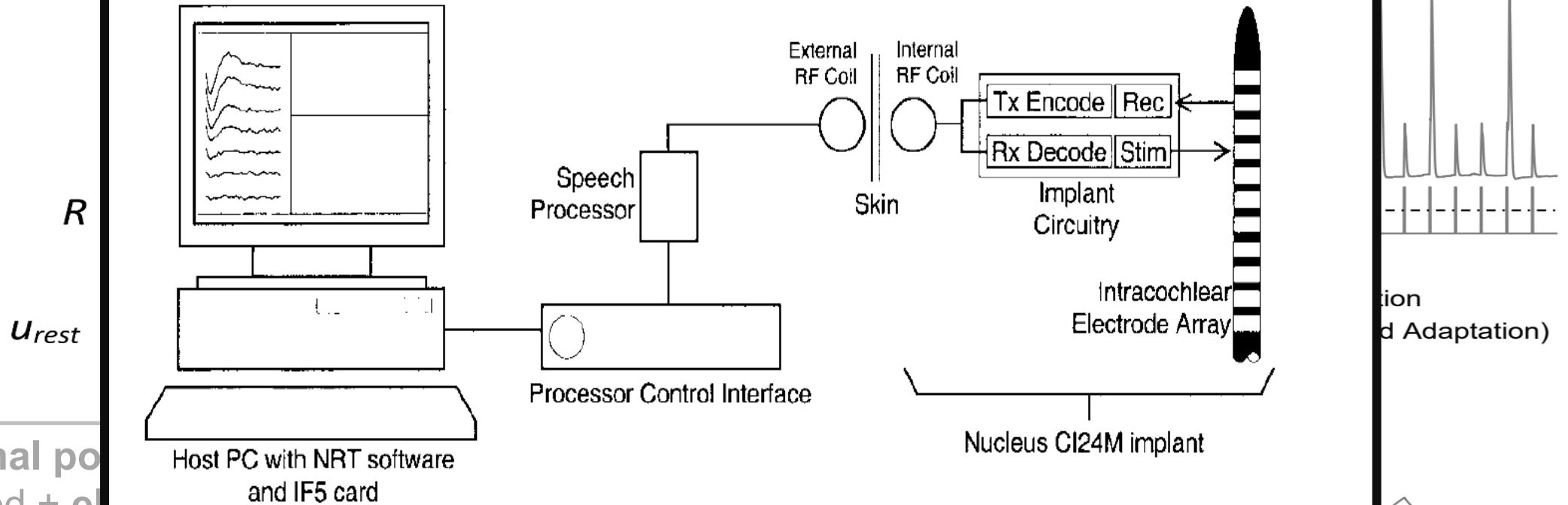
D Accommodation (Subthreshold Adaptation)



Model: Integrate & Fire model with dynamic threshold + ECAP model

Measurable data = sum of action potentials of all neurons (ECAP)

To enable fitting to data, we also model the ECAP signal generation.



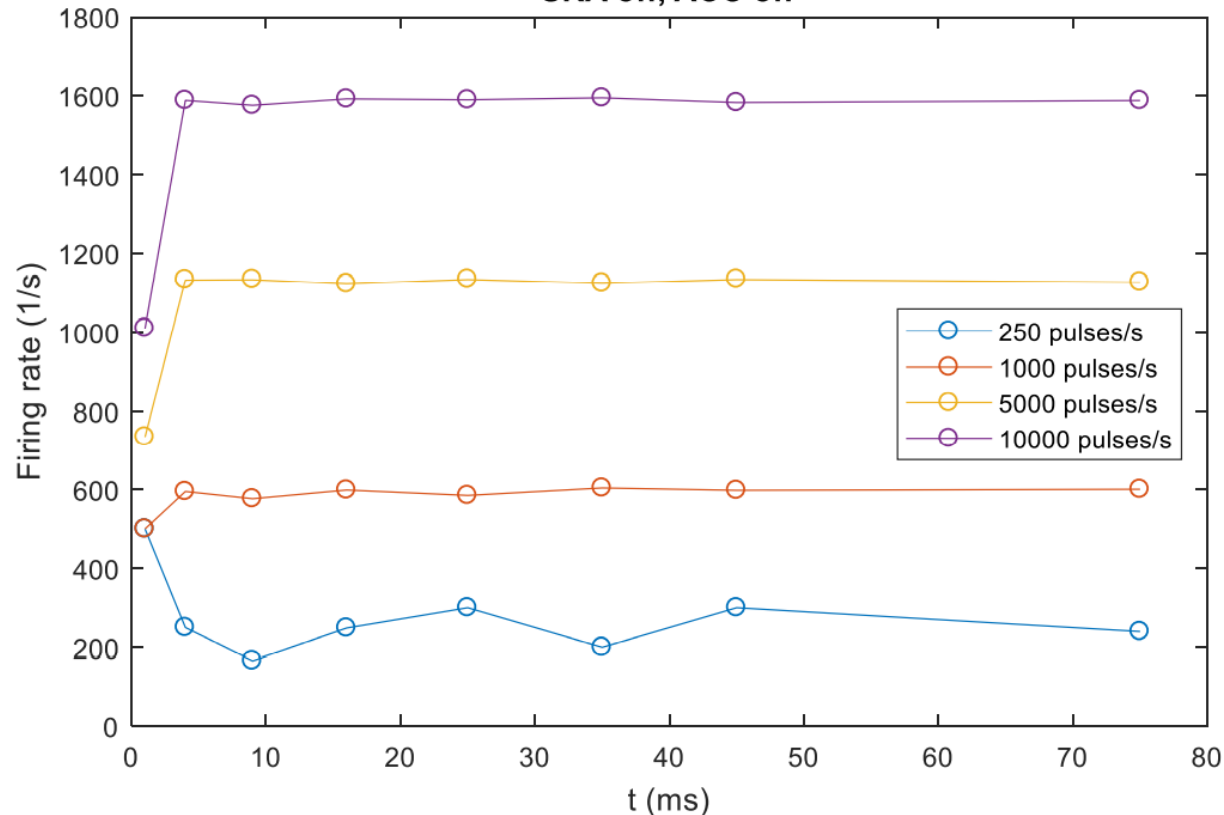




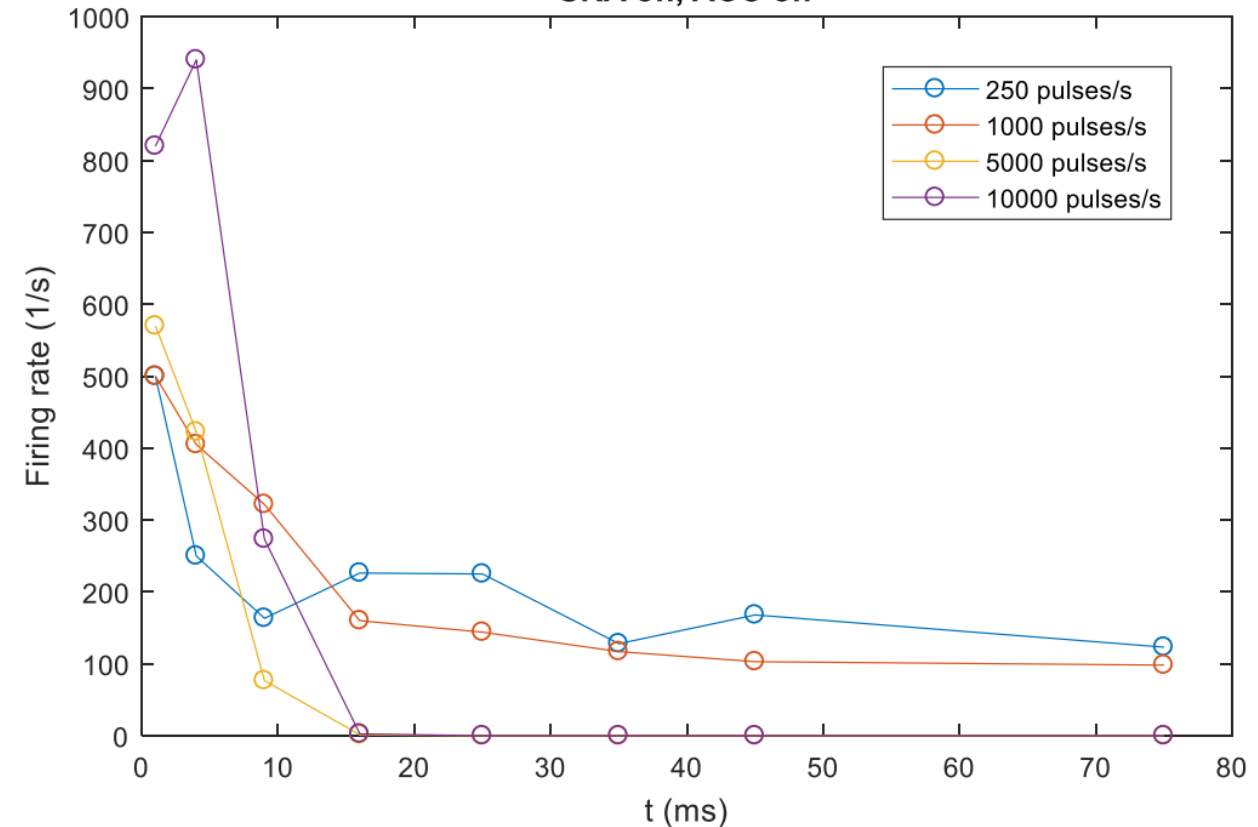
Neuronal population response for pulse trains of various frequencies

Spike-rate adaptation (SRA) strongly influences the response!

Firing rate vs. time, various IPI
neurons: 100, Istim = 62 pA
SRA off, ACC off

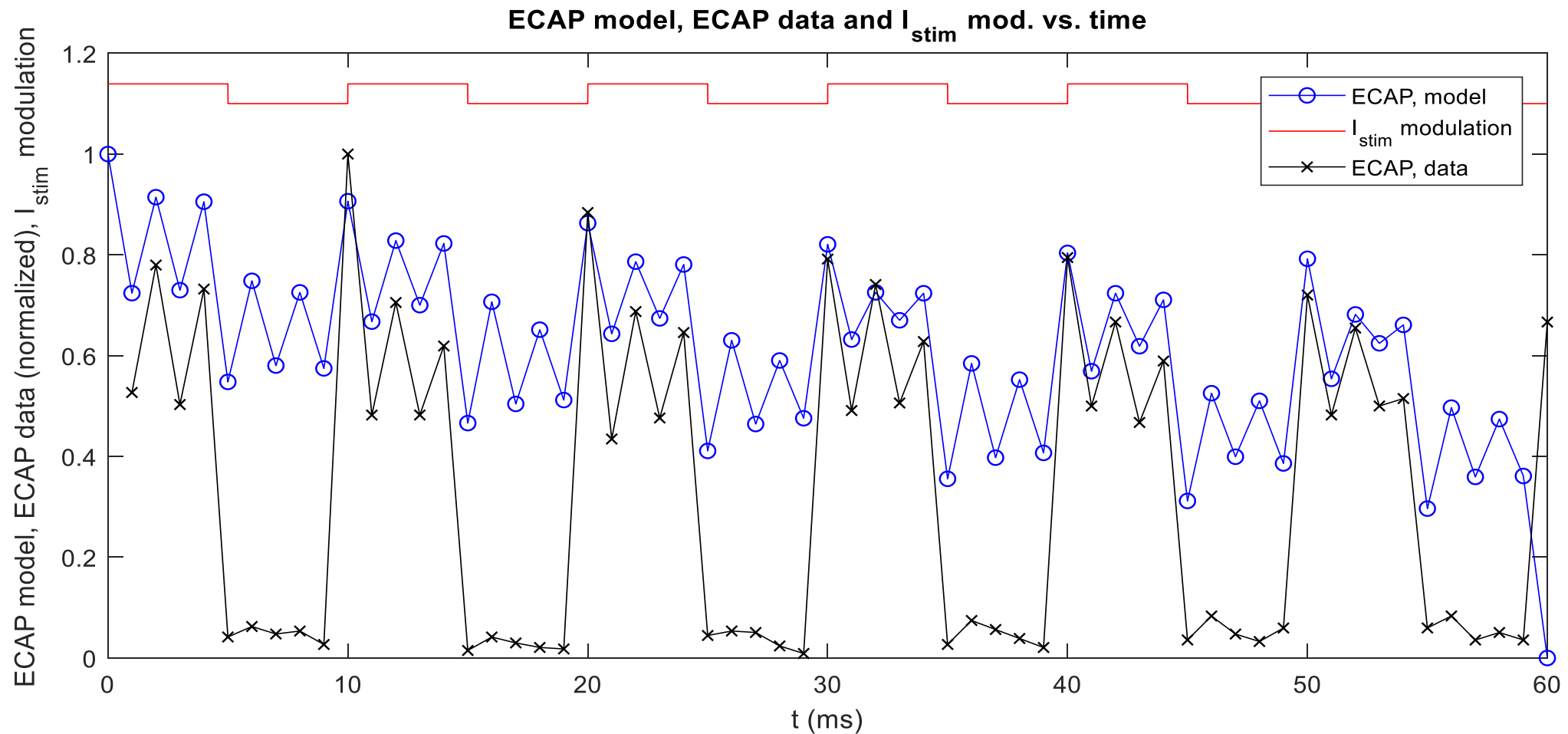


Firing rate vs. time, various IPI
neurons: 100, Istim = 62 pA
SRA on, ACC on



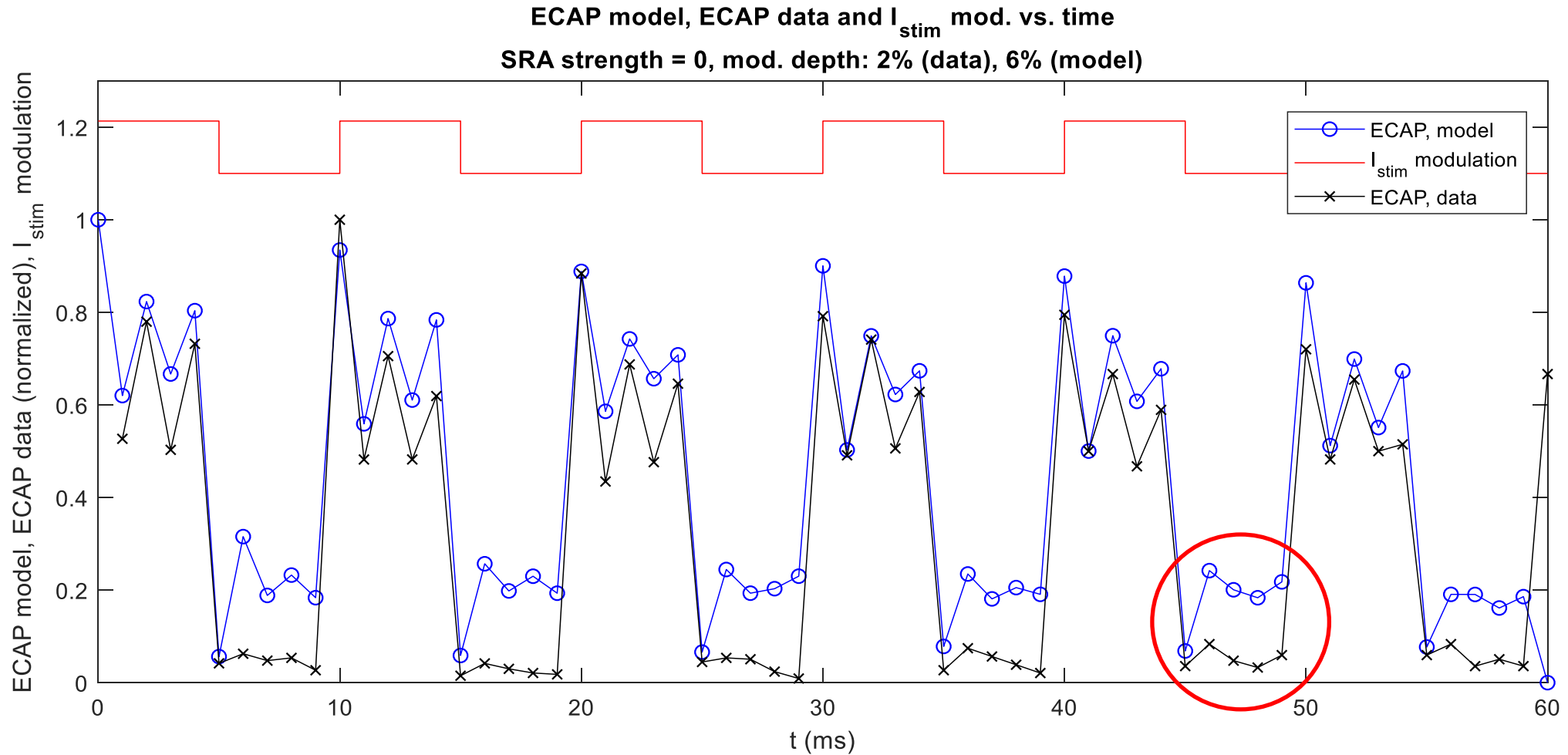


Square modulated pulse train: no fitting of model parameters



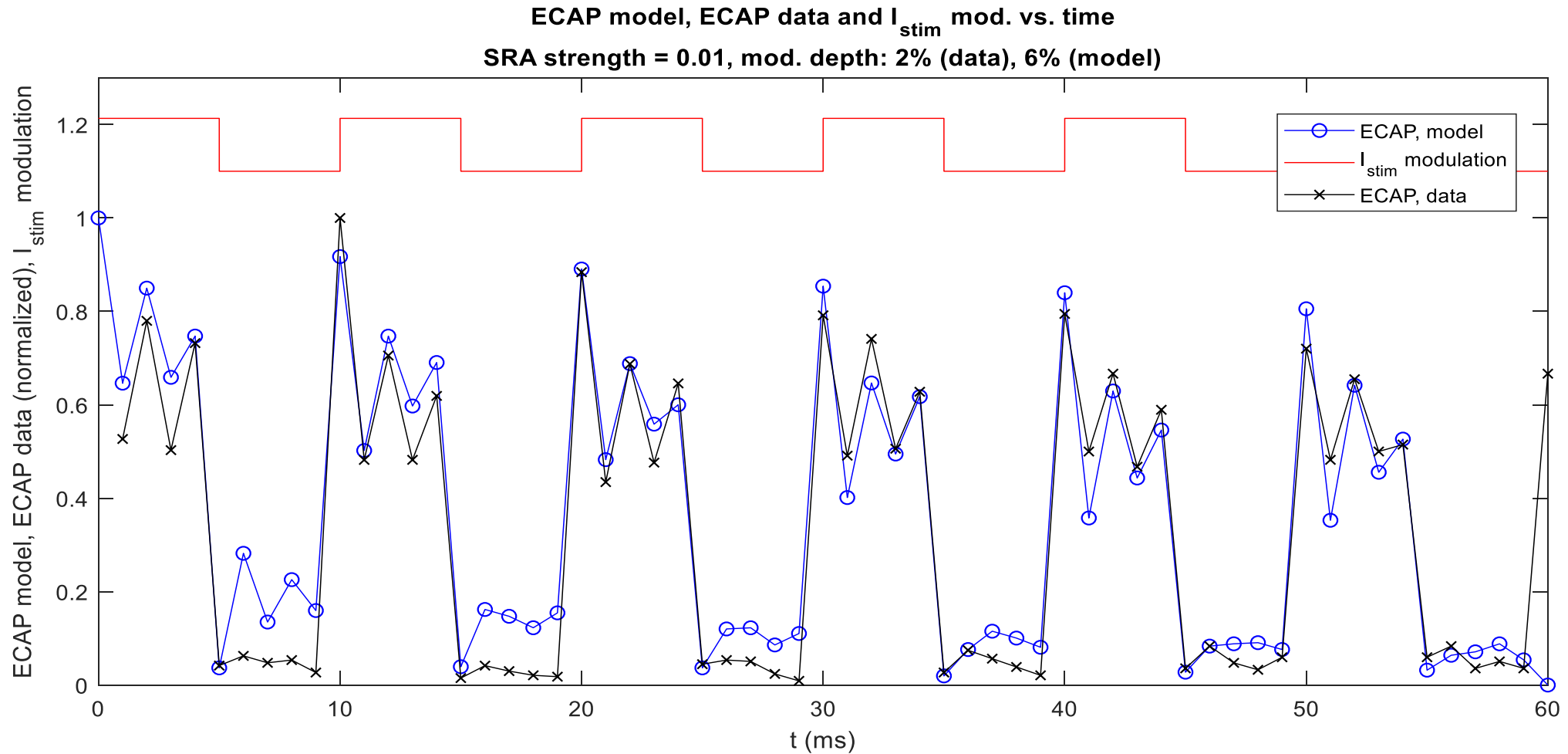


Square modulated pulse train: parameters fitted, no SRA





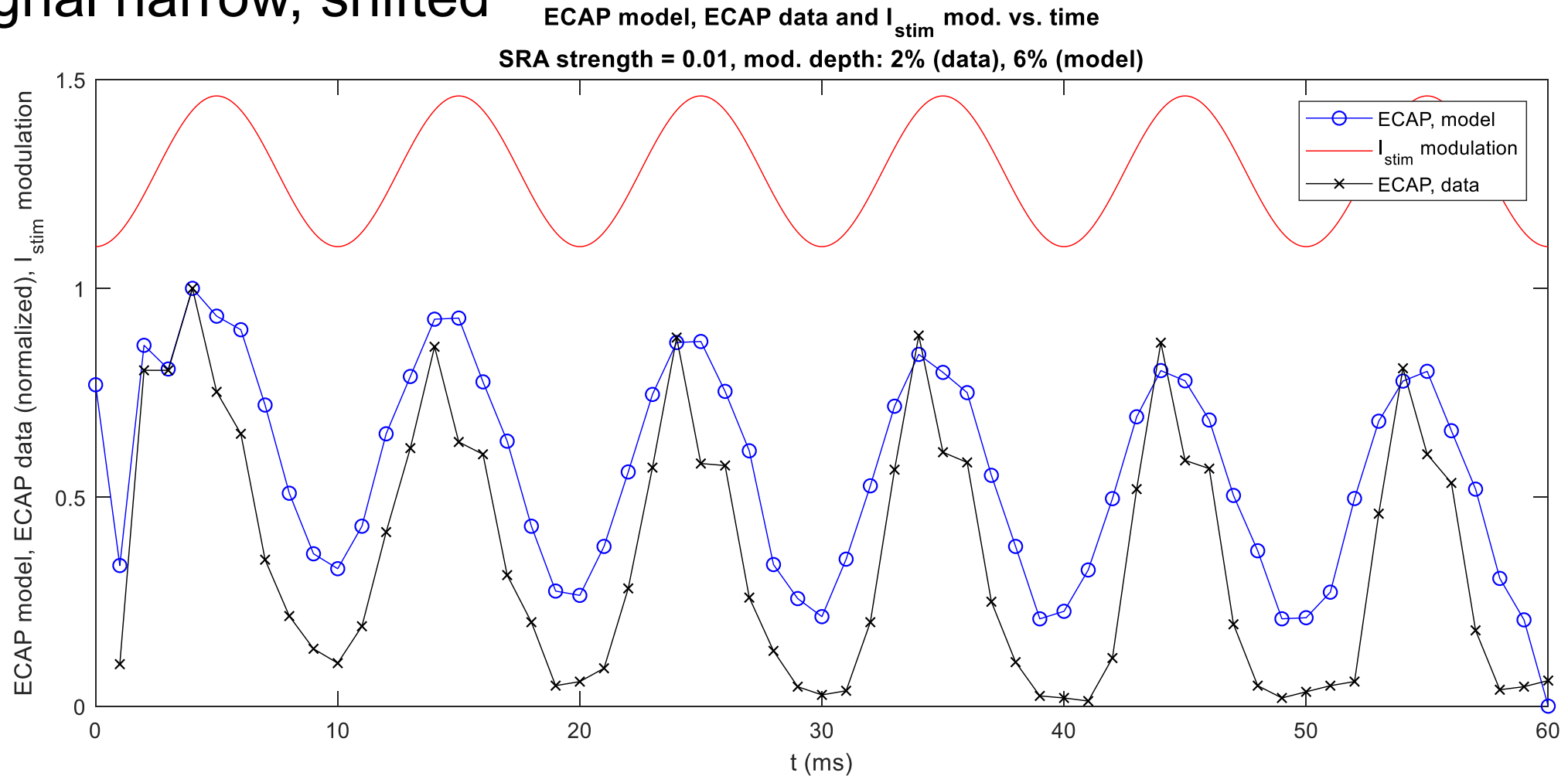
Square modulated pulse train: $SRA = 0.01 \rightarrow$ better fit

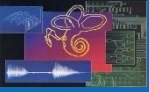




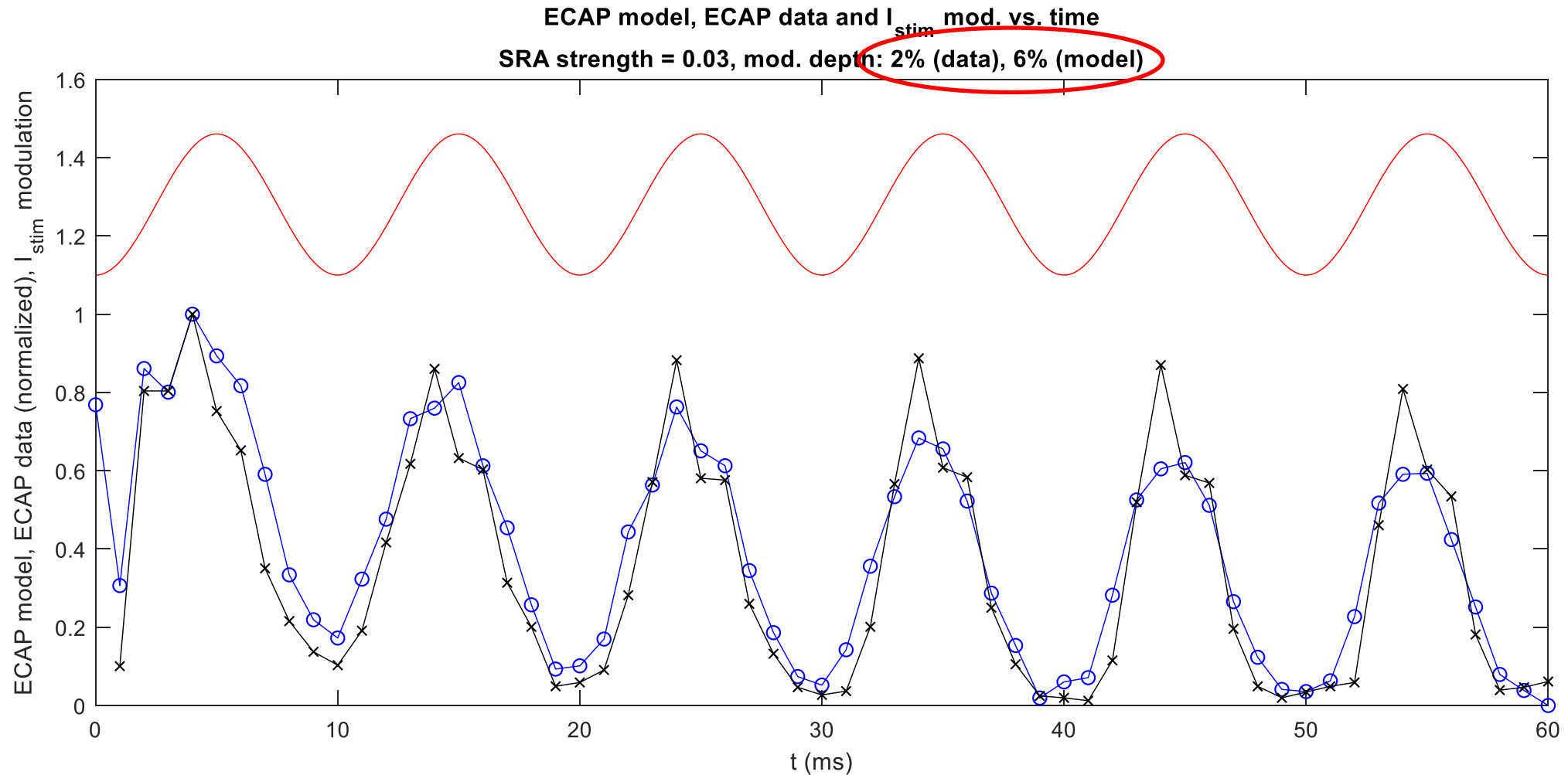
Sinusoidally modulated pulse train: no SRA

→ signal narrow, shifted



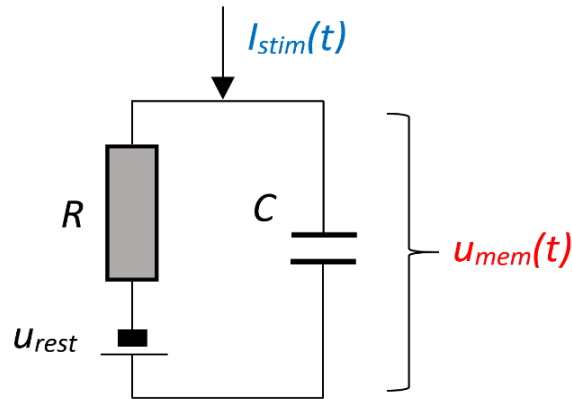


Sinusoidally modulated pulse train: $SRA = 0.03 \rightarrow$ much better fit



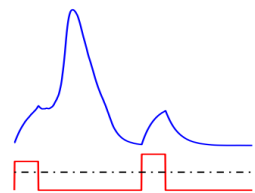
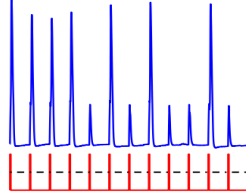
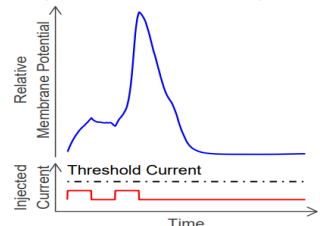
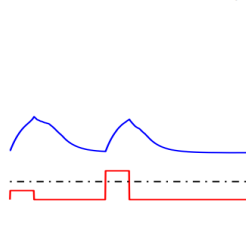


Conclusion, further work



- Single neuron I&F model with dynamic threshold extended to a neuronal population, ECAP model included.
- Our model can predict shape of the neural responses for various modulated pulse trains...
- ...but **inadequate ECAP model** (low response vs. stimulus slope) leads to **underestimated ECAP response modulation depths**.
- **Remaining challenges:** developing a more adequate ECAP model (e.g. additional time delays in single neuron responses), comparison and fitting with more data, modelling negative pulses.
- **Subsequent publication:** *Tabibi, et al. A bio-inspired coding (BIC) strategy for cochlear implants, Hearing Research, 2020*

A Refractoriness

B Spike-Rate Adaptation
(Spike-Dependent)C Facilitation
(Temporal Summation)D Accommodation
(Subthreshold Adaptation)

Boulet J. , *Biophysical and phenomenological models of cochlear implant stimulation*, PhD Thesis McMaster University 2016



Thank you for your attention!

Questions?



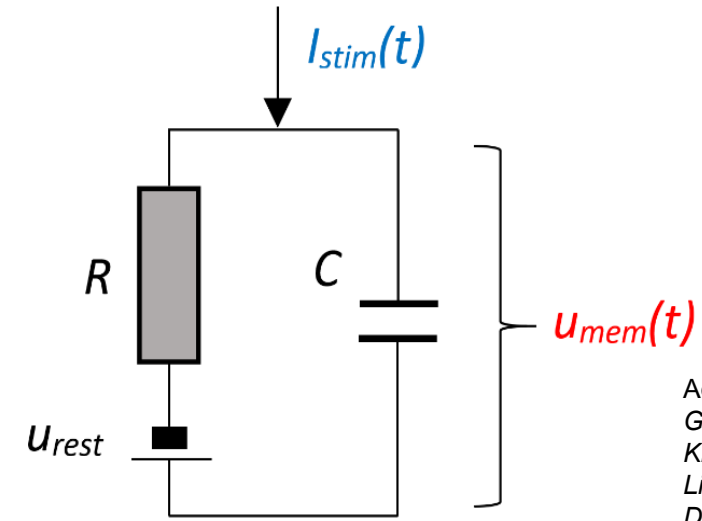
Leaky neuronal membrane

- Insulating membrane → capacitor
- But slightly conductive → resistor

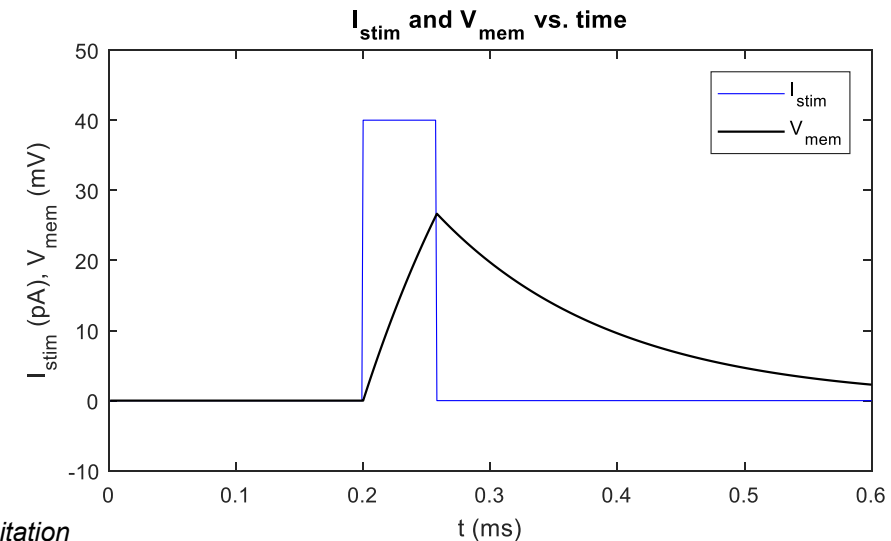
$$I_{stim}(t) = I_c(t) + I_R(t)$$

$$\frac{dv}{dt} = -\frac{v(t)}{RC} + \frac{I_{stim}(t)}{C}$$

- Behavior of the system:
 - discharges with time scale $RC \approx 0.13 \text{ ms}$ [ref]
 - Charges linearly for short pulses, and tends to saturate for longer pulses



Adapted from Wulfram Gerstner, Werner M. Kistler, Richard Naud and Liam Paninski, *Neuronal Dynamics*, Cambridge University Press 2014



Lapicque L, *Recherches quantitatives sur l'excitation électrique des nerfs traitée comme une polarisation*, 1907, *J Physiol Pathol Gen Paris* 9:620–635



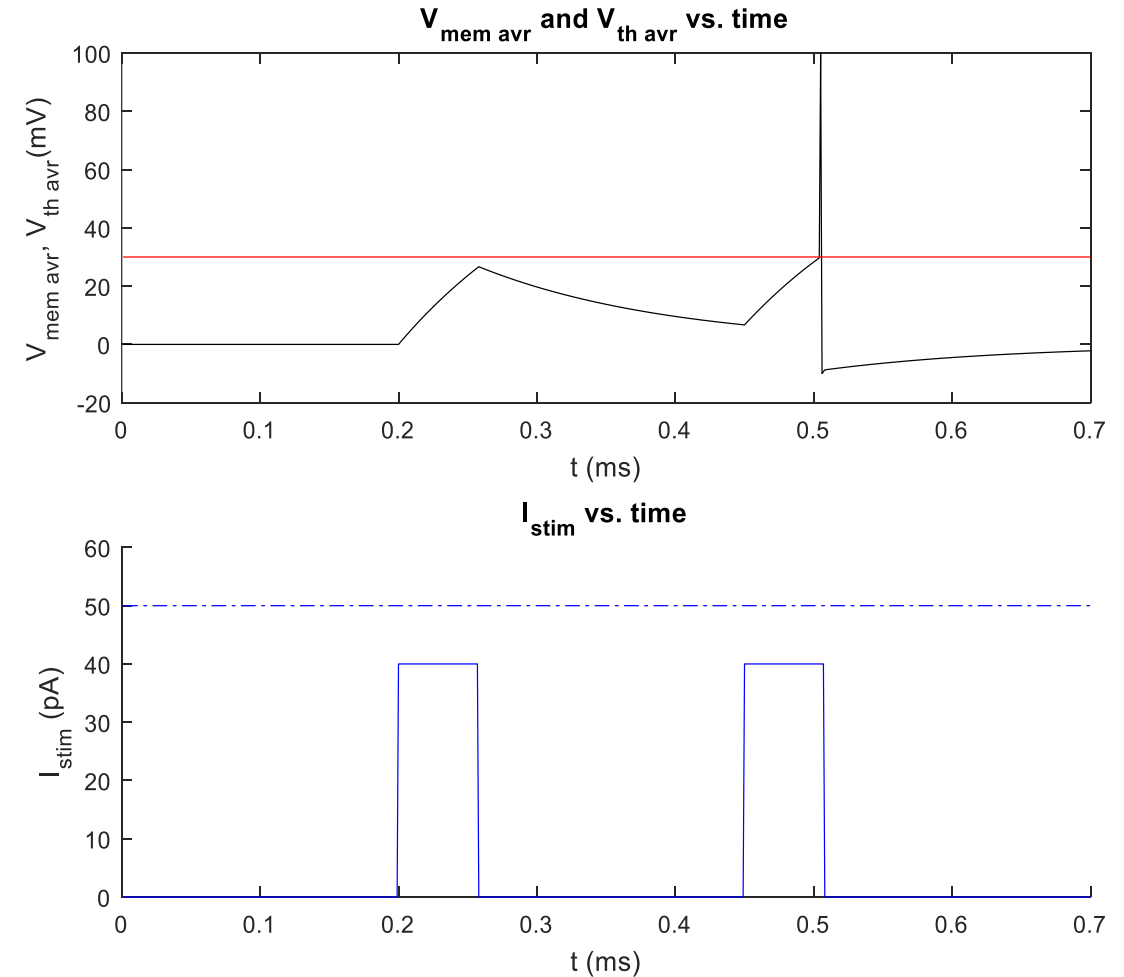
Integrate and fire model: RC + spike when above threshold

- If we now make the neuron fire whenever:

$$v \geq v_{threshold}$$

we have the *integrate and fire model (IF)*

- Simplest neuronal model (which produces spikes), yet able to predict timing of spikes for low stimulus frequencies (ca. <500 Hz)





Noise

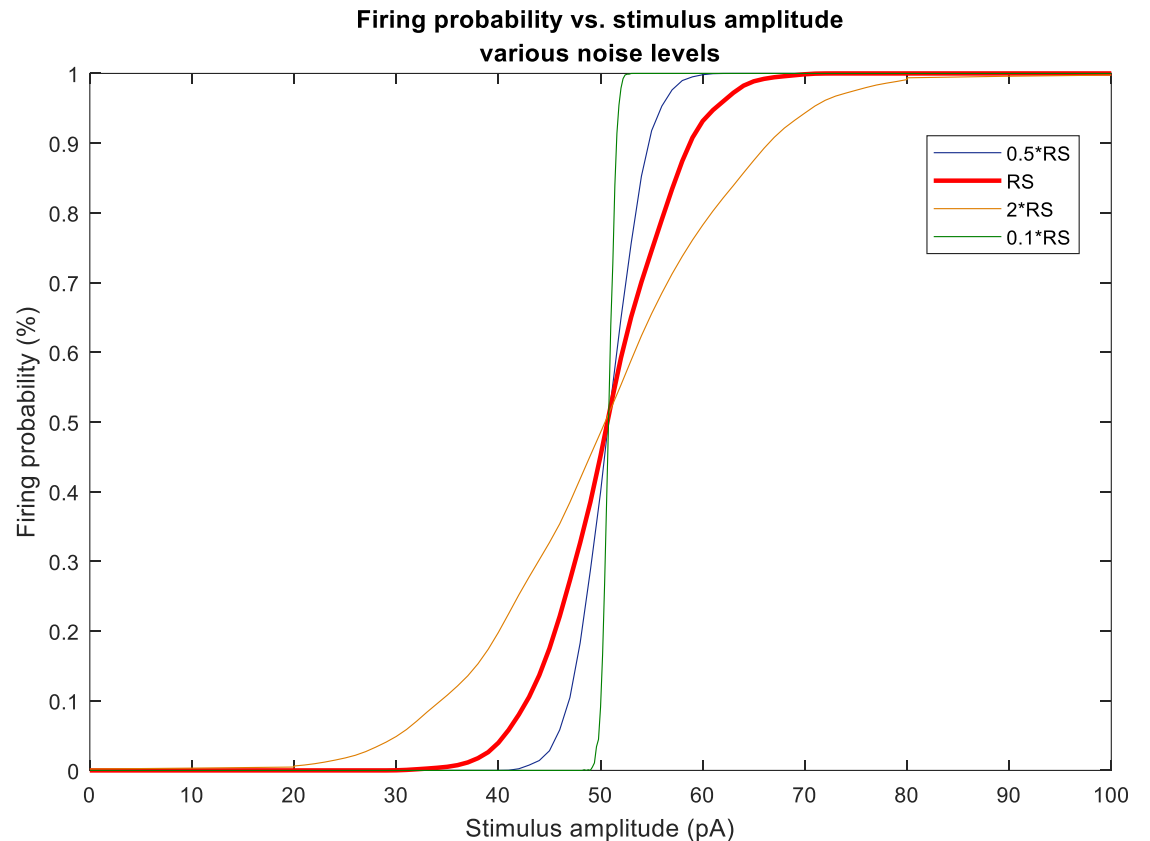
- Noise makes the threshold less sharp and reduces synchronization in neural response.
- Sources: random fluctuations in ion concentrations, changes in ion channel conformation (conductivity).
- Noise used in this model is:
 - added to the threshold voltage
 - normally distributed in amplitude
 - lowpass filtered (200 Hz cutoff) [ref Fredelake]
 - has standard deviation equal to:

$$v_{noise\ std} = RS * v_{threshold}$$

where:

$$RS = 0.05$$

- RS is scaled by each neuronal phenomena!



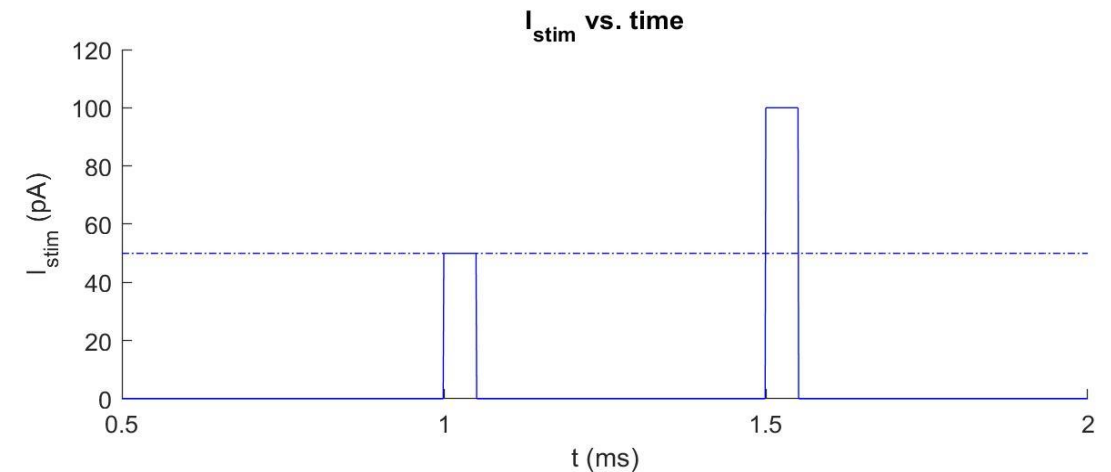
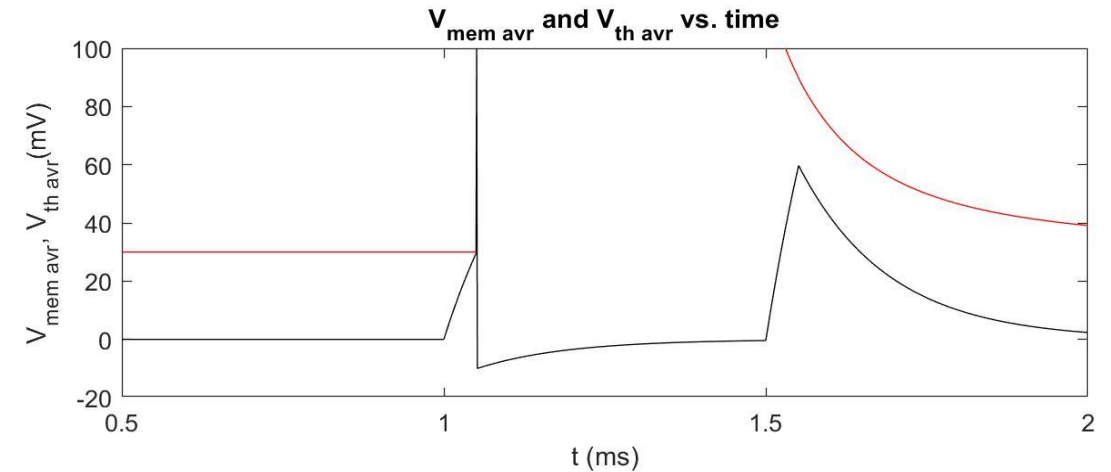
Verveen A. A., Derksen, H. E., *Fluctuation phenomena in nerve membrane*, 1968, *Proc. IEEE* 56, 906–16.

Boulet J., *Biophysical and phenomenological models of cochlear implant stimulation*, 2016, *PhD Thesis McMaster University*



Refractoriness → „typical“ neuronal behavior 1

- When neuron spikes, for a short period it will be nonexcitable (ARP), and then it will slowly return to its rest excitability (RRP).
- Caused by Na^+ channel inactivation after spike [ref HH], ARP/RRP prolonged by HCN and KLT channels.
- Modelled by multiplying the firing threshold with $x_{th\ ref}$, which is:
 - infinite while in ARP ($t \leq T_{ARP} = 0.332\text{ ms}$)
 - exponentially decreasing during RRP ($t > T_{ARP}$) with time constant $\tau_{RRP} = 0.411\text{ ms}$

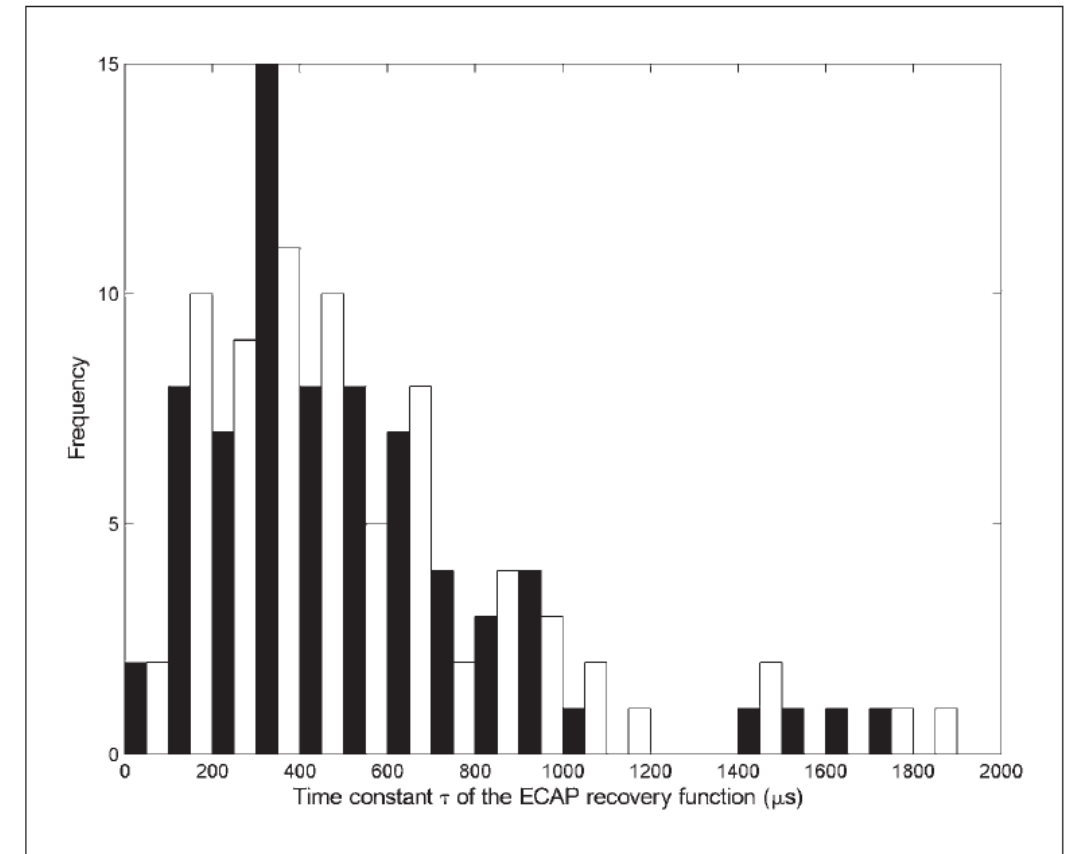


Negm M.H., Bruce I.C., The effects of HCN and KLT ion channels on adaptation and refractoriness in a stochastic auditory nerve model. *IEEE Trans Biomed Eng.* 2014 Nov; 61(11):2749-59



Refractoriness → „typical“ neuronal behavior 2

- T_{ARP} and τ_{RRP} show great variation in and between patients, which is caused by variations in number of related ion channels.
- Therefore a normal distribution around mean values is considered, with variations being correlated.
- This causes reduction in response oscillations, especially for high frequency pulses, and increases the difference between response to the first pulse in the train and the remaining pulses.



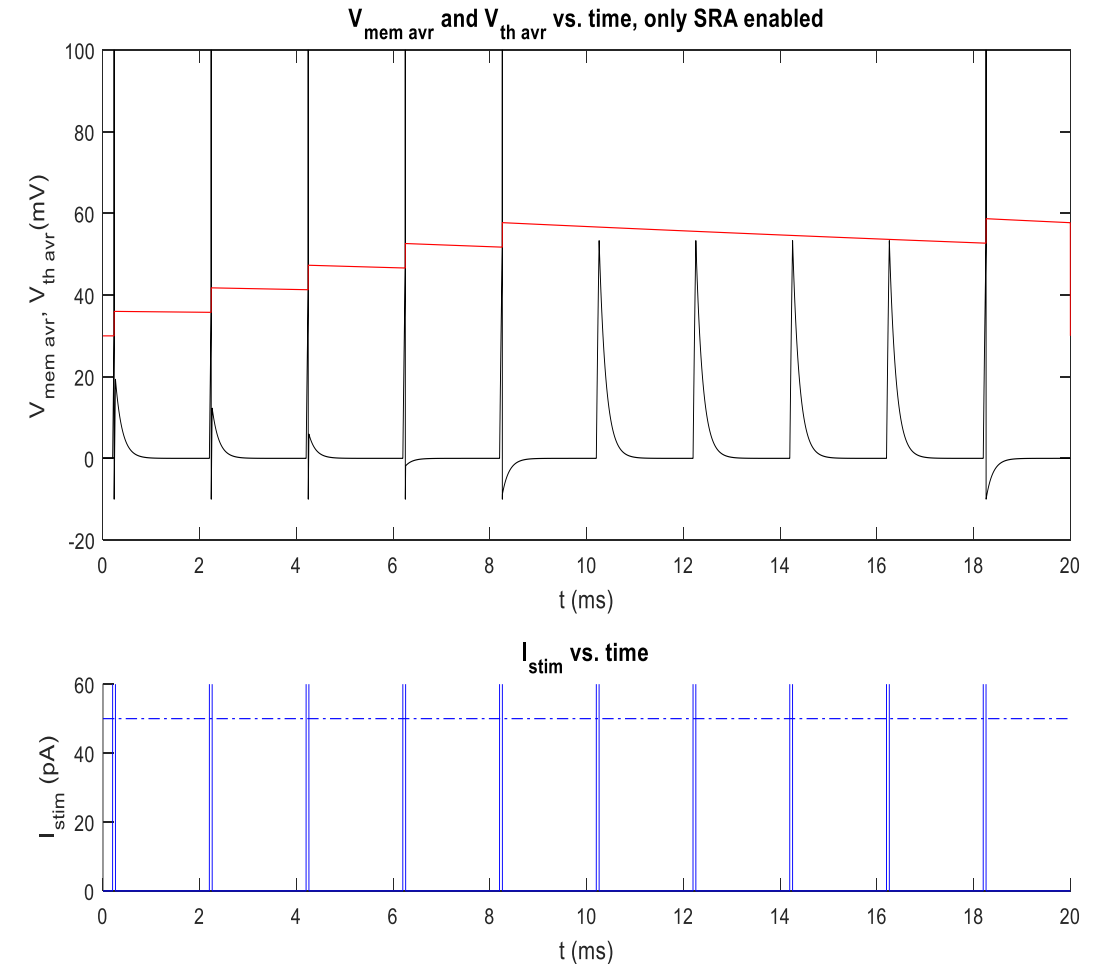
Rubinstein J.T., Pseudospontaneous activity: stochastic independence of auditory nerve fibers with electrical stimulation, *Hearing Research* 1999, 127 108-118

Adapted from Morsnowski et al., *Measuring the Refractoriness of the Electrically Stimulated Auditory Nerve*, *Audiol Neurotol* 2006;11:389–402



Spike-rate adaptation

- „Spiking neurons pay a high metabolic cost to encode information, hence neurons will tend to reduce their firing rate“.
- Caused by increase in extracellular K^+ conc. due to K^+ outflux at each spike, activation of HCN channels etc.
- Modelled with a factor $x_{th\ SRA}$ which:
 - „jumps up“ by $p_{th\ SRA} = 0.04$ at each spike
 - exponentially decreases to 1 with $\tau_{th\ SRA} = 50\ ms$
- Influence on noise is assumed to be identical.

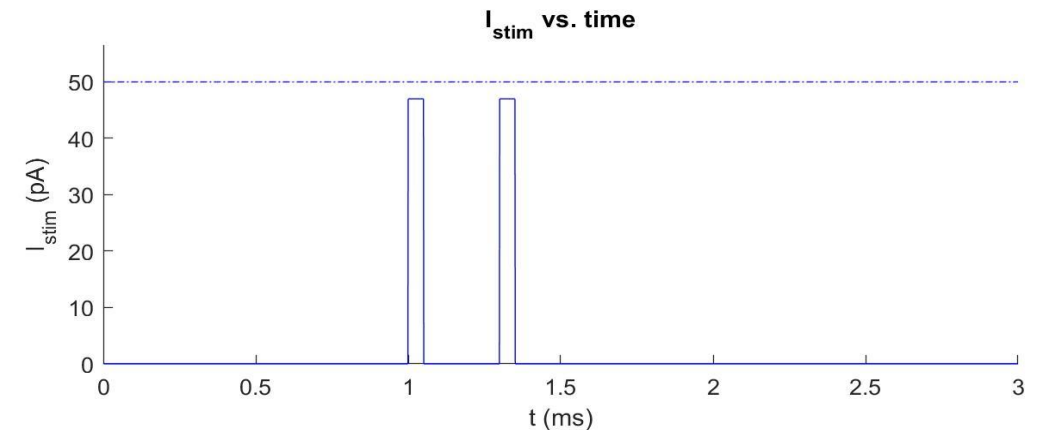
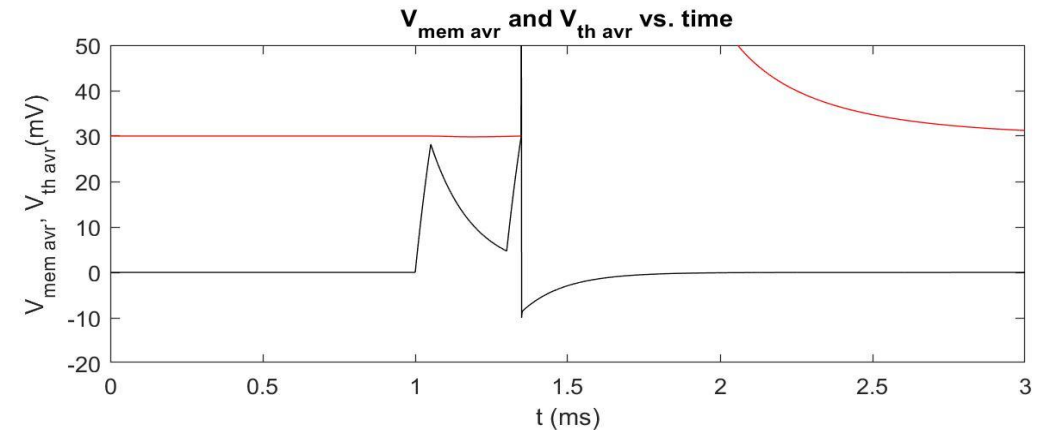




Facilitation and accommodation

→ subthreshold phenomena, typically ignored but important!

- (Active) facilitation:
 - First pulse makes the neuron more excitable, active during timescale of 0.5 ms or less.
 - Caused by activation of Na^+ channels, making membrane more conductive to Na^+ ions.
- Accommodation:
 - Similar but opposite to facilitation, overpowers facilitation after 0.5 ms and last for ca. 1 ms.
 - Caused by inactivation of Na^+ channels, KLT channel activation etc.





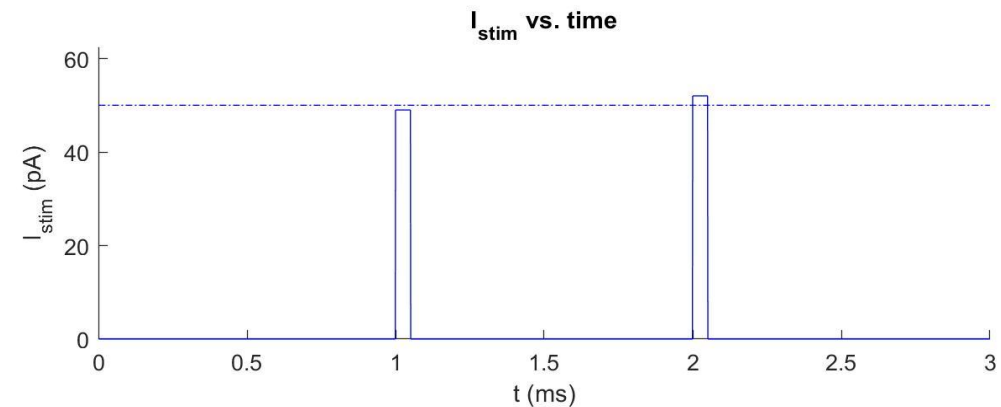
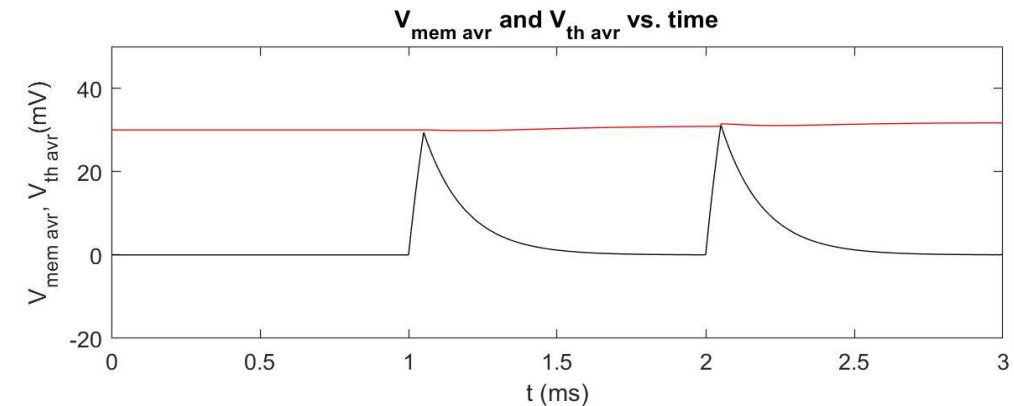
Facilitation and accommodation

- Key difference from other phenomena is that facilitation and accommodation depend on the stimulus level.
- Both phenomena modelled as:

$$\frac{dy_{thf,a}}{dt} = -\frac{y_{thf,a}}{\tau_{thf,a}} + a_{thf,a} z(t)$$

where $z(t)$ is the normalized membrane voltage (relative to threshold).

- Finally, we have: $x_{thf,a} = 1 + y_{thf,a}$

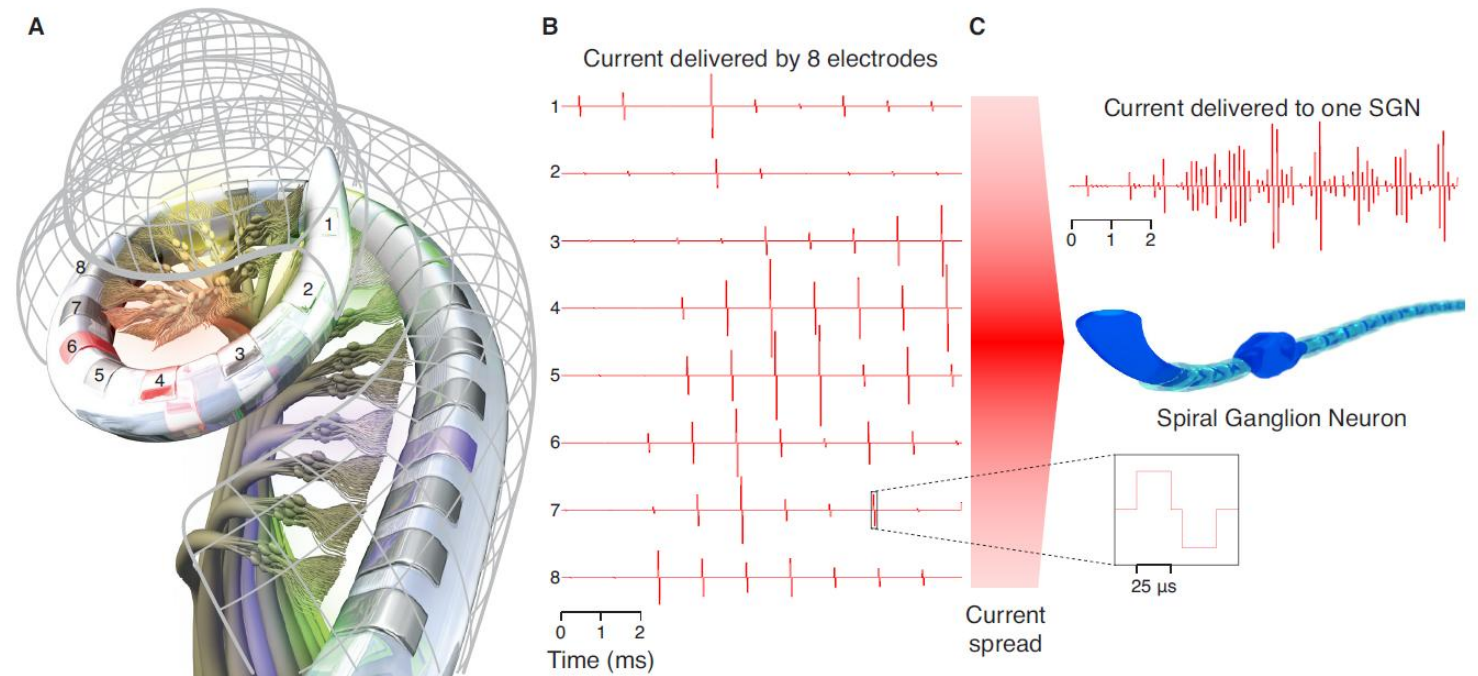




Spread of excitation: across the neurons, and extra- to intra-cellular

- Electrodes are in a conductive medium, causing stimulus spread (both good and bad news).
- Longitudinal spread is typically modelled as exponential-absolute function (spread width ca. 0.8 mm)
- How much current actually reaches the neuron is hard to measure, and thus calculated as a fixed scaling – ratio of *measured extracell. current threshold*, and *model-obtained intracell. threshold for fixed pulse width*.

But how do you measure spread...or anything?

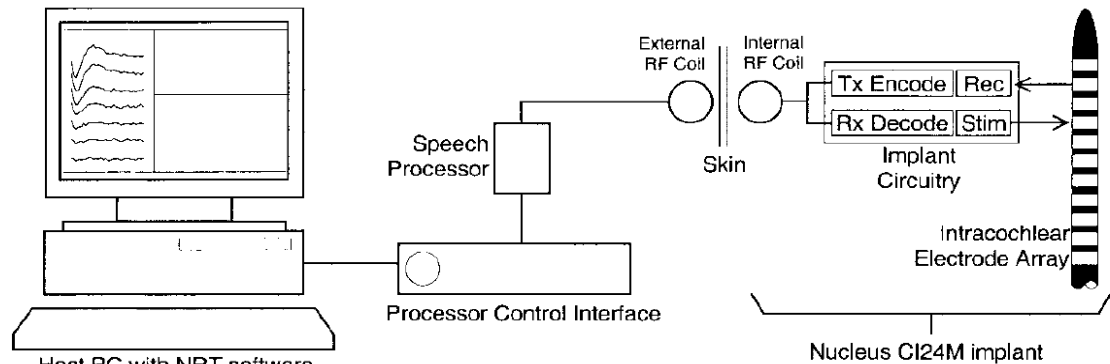


Adapted from Boulet J., *Biophysical and phenomenological models of cochlear implant stimulation*, PhD Thesis McMaster University 2016

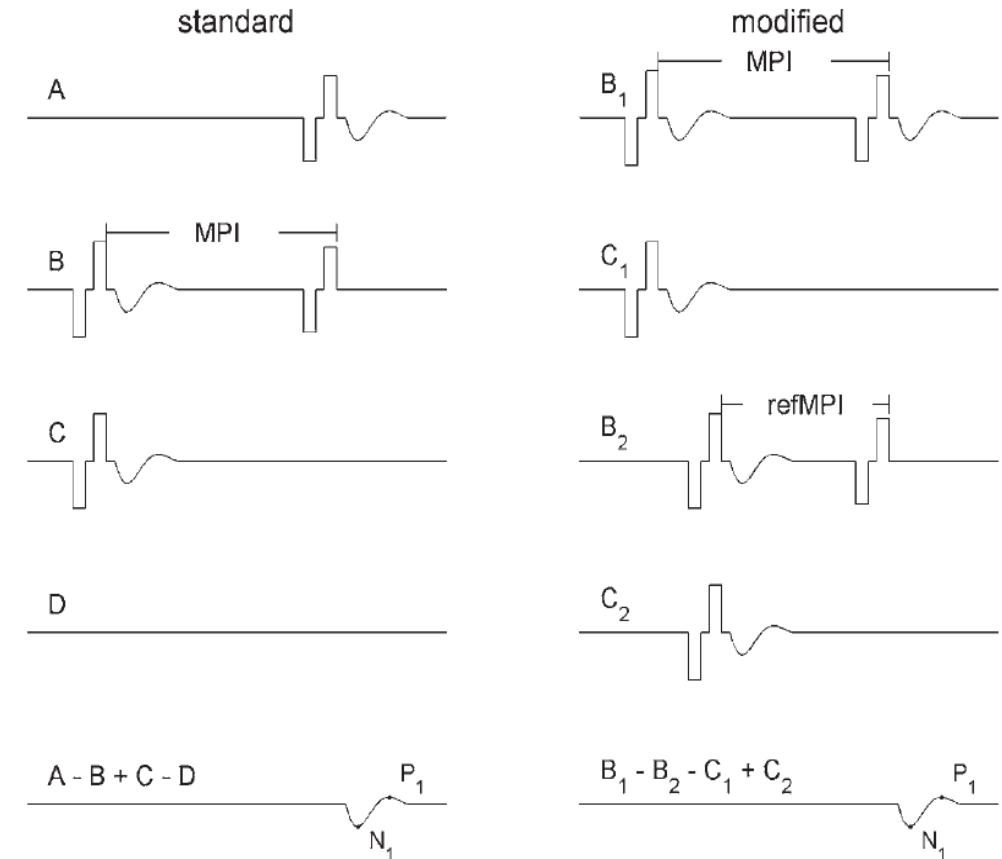


ECAP: measuring the neuronal response

- Each spiking neuron produces an action potential (spike), and a sum of these can be recorded by CI electrode.
- Signal depends on how many neurons are present (have survived!), how excitable they are, and on the latencies of neural APs.
- To remove the probe/masker artifact, subtraction measurement method is used.



Dillier N., Lai W.K., Measurement of the electrically evoked compound action potential via a neural response telemetry system, *Ann Otol Rhinol Laryngol* 111:2002



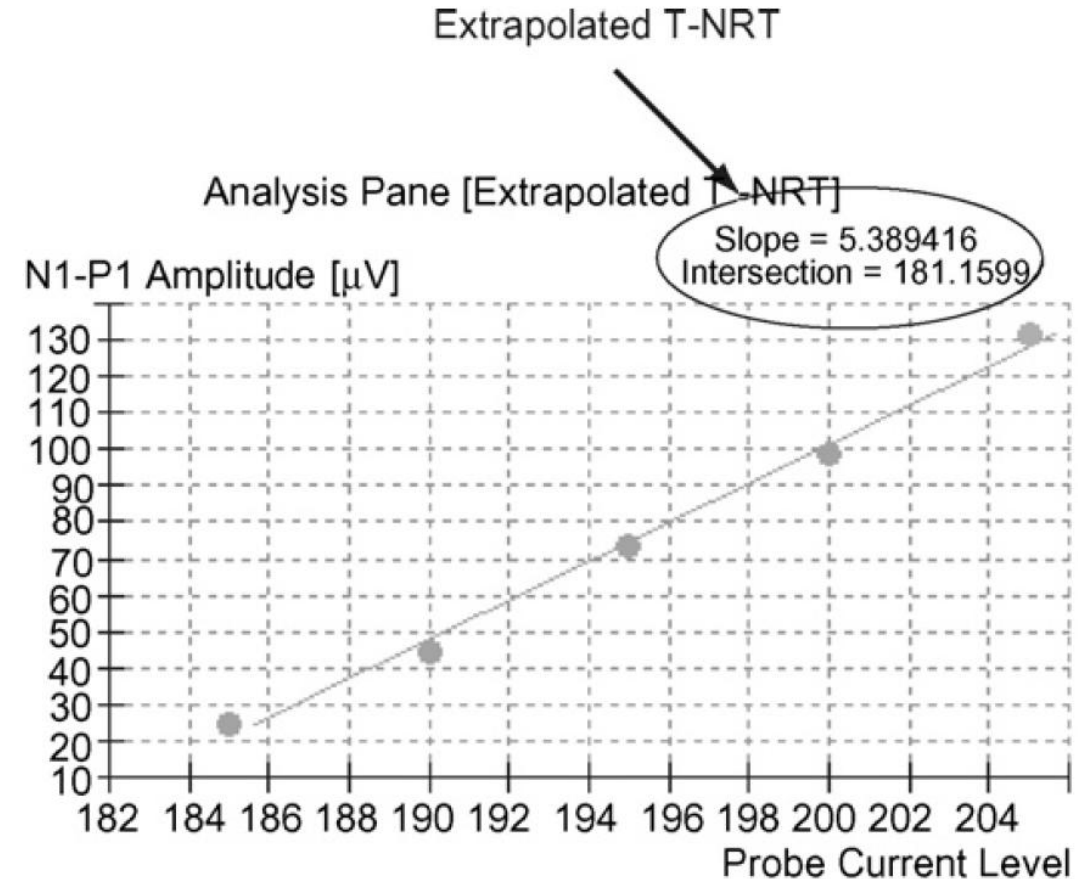
Miller C.A., Abbas P.J., Brown C.J., An improved method of reducing stimulus artifact in the electrically evoked whole-nerve potential, *Ear Hear.* 2000 Aug; 21(4) : 280-90.



ECAP: amplitude growth function 1

- How does the measured signal grow with stimulus level?
- Determined by applying a single pulse at a various stimulus level and measuring the ECAP response.
- If stimulus is measured in μA this relation is logarithmic – for convenience a unit called Current Level (CL) is used, where:

$$I = 17.5 \mu\text{A} * 100^{CL/225}$$

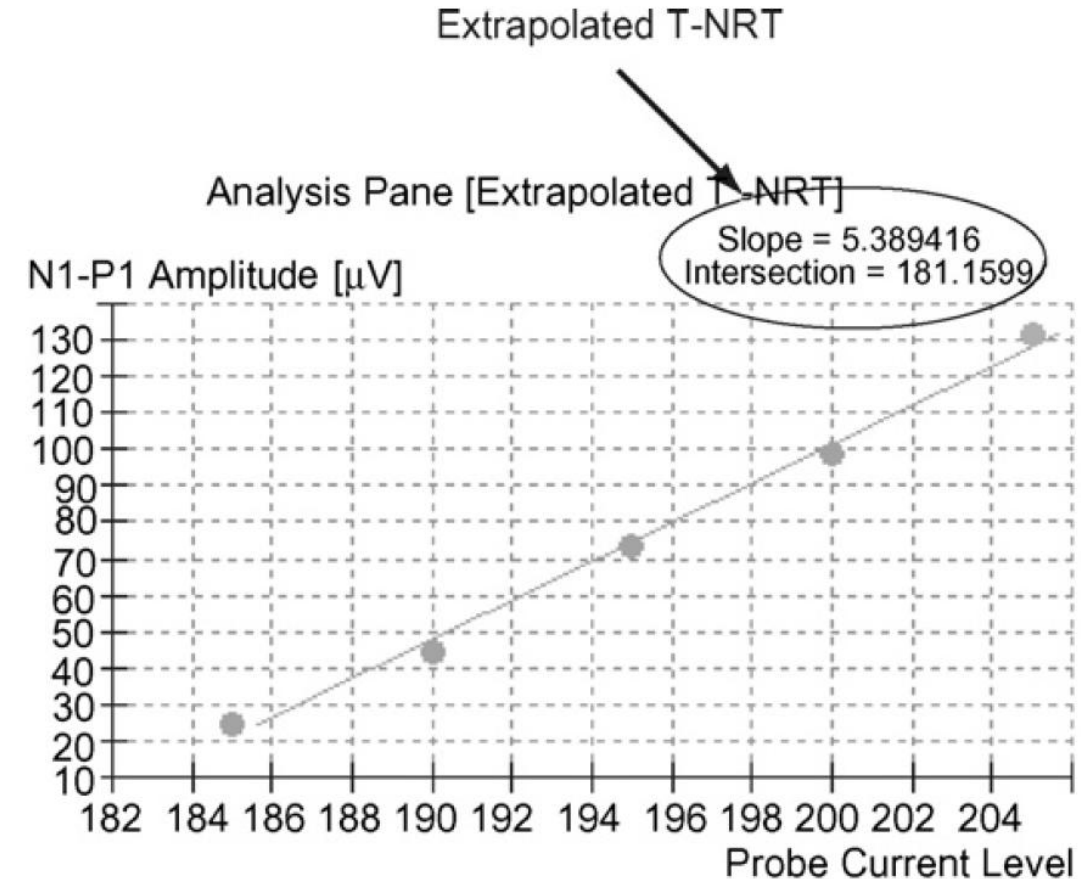


Botros et al., AutoNRT: An automated system that measures ECAP thresholds with the Nucleus Freedom cochlear implant via machine intelligence, *Artificial Intelligence in Medicine*, 40, 15-28, 2017.



ECAP: amplitude growth function 2

- Which parameters can be obtained from AGF?
 - Threshold CL level → extracell.-to-intracell. current spread scaling (also obtainable from subjective threshold!)
 - ECAP amplitude and AGF slope → number of neurons activated as stimulus increases → number of surviving neurons (assuming no latency)
- Note: Stimulus used in AGF spreads to other neurons, hence parameters above are not local but population averages weighted by the spread-of-excitation function.



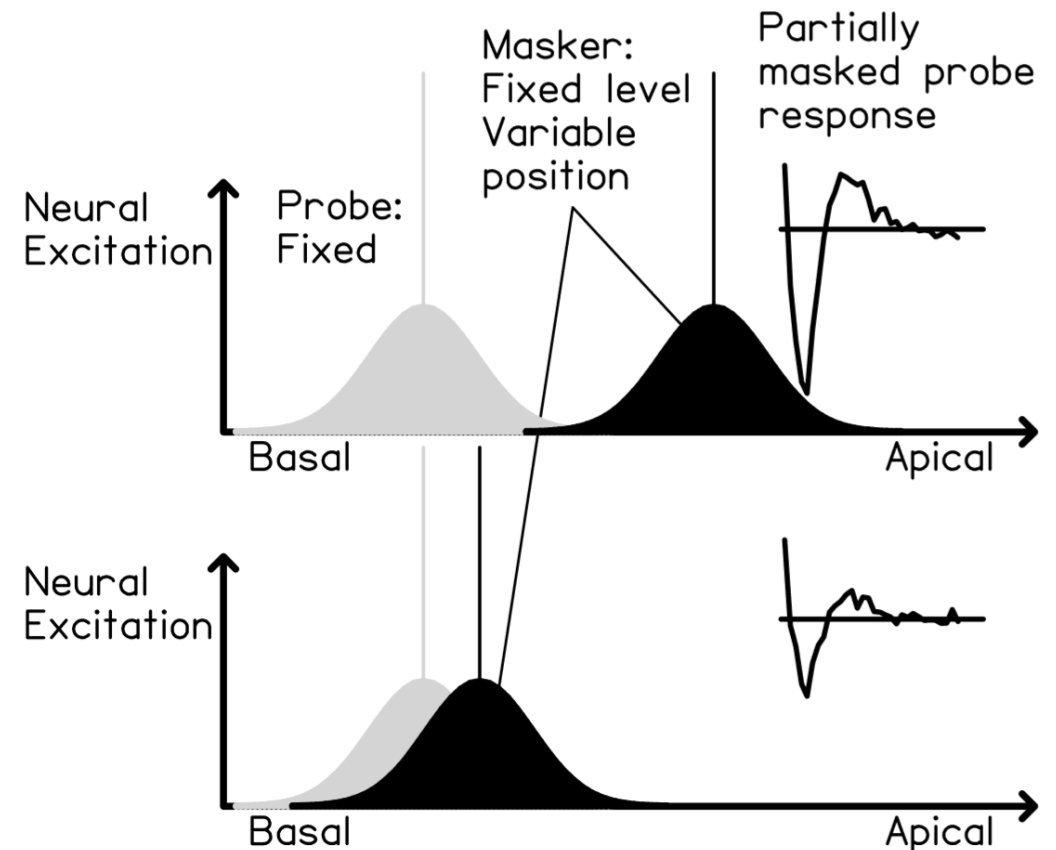
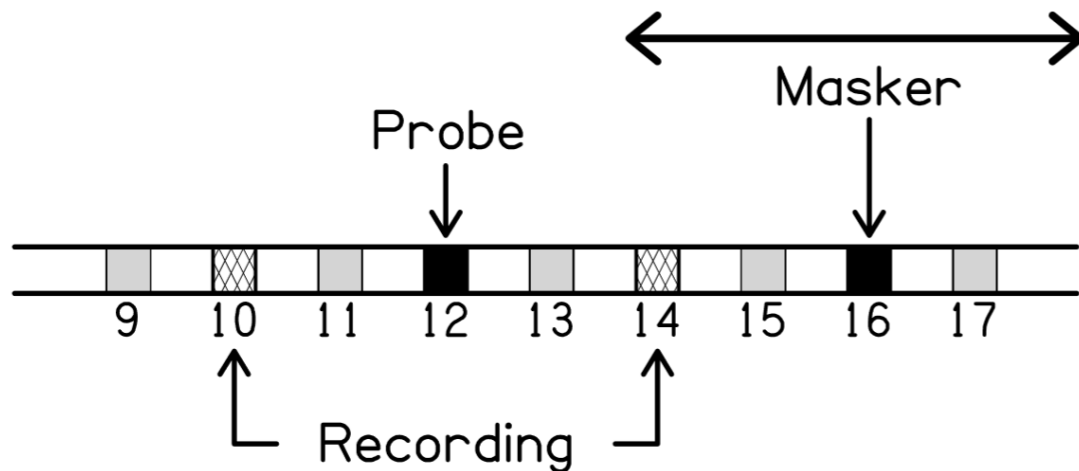
Botros et al., AutoNRT: An automated system that measures ECAP thresholds with the Nucleus Freedom cochlear implant via machine intelligence, *Artificial Intelligence in Medicine*, 40, 15-28, 2017.

What if we use two excitation pulses?

Pulses at two different electrodes

→ (2nd pulse) ECAP signal proportional to remaining excitability of neurons

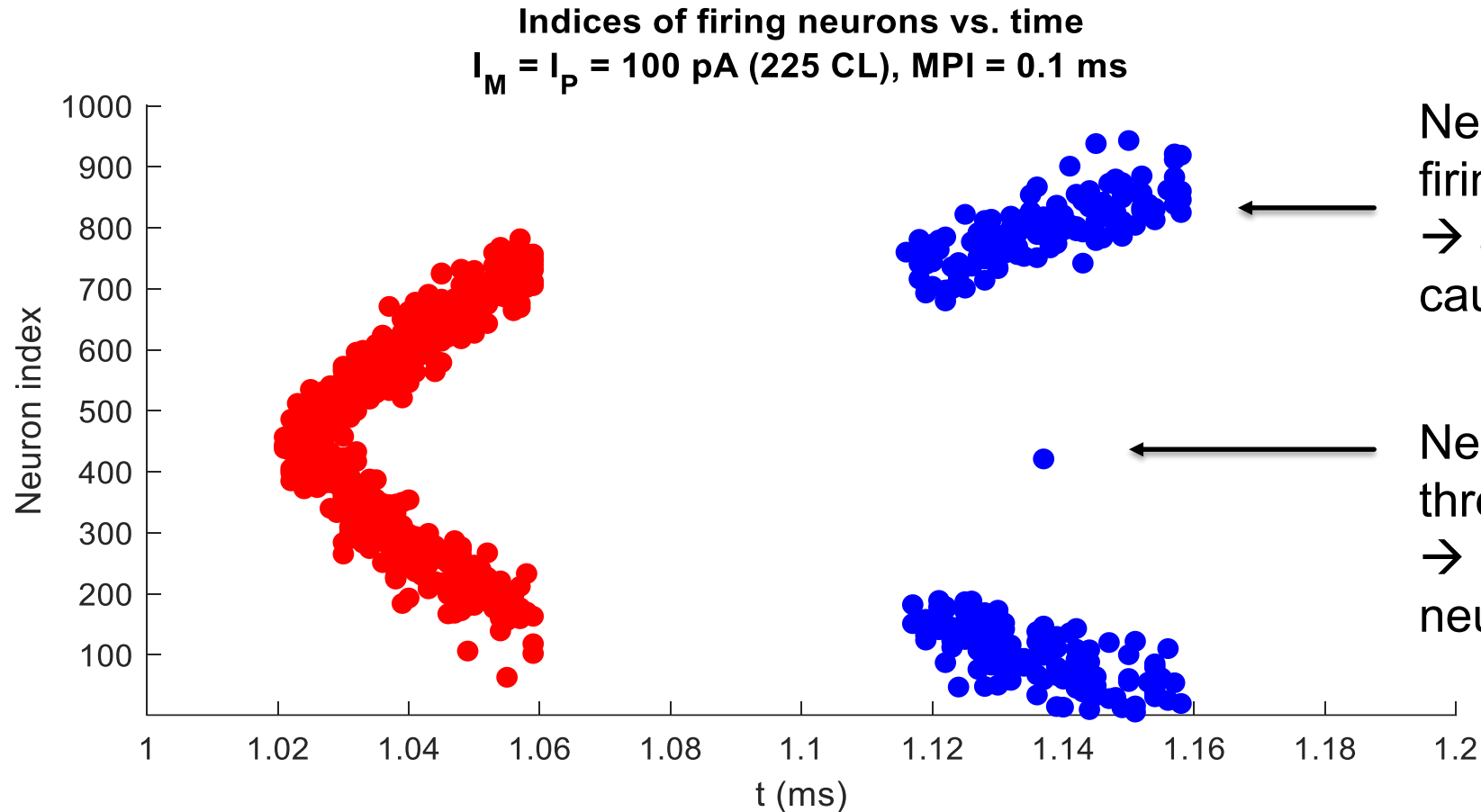
→ we have a measure of spread of stimulus excitation





Two excitation pulses with a time delay (but at same location)?

→ again a measure of remaining excitability - depends on refractoriness etc.

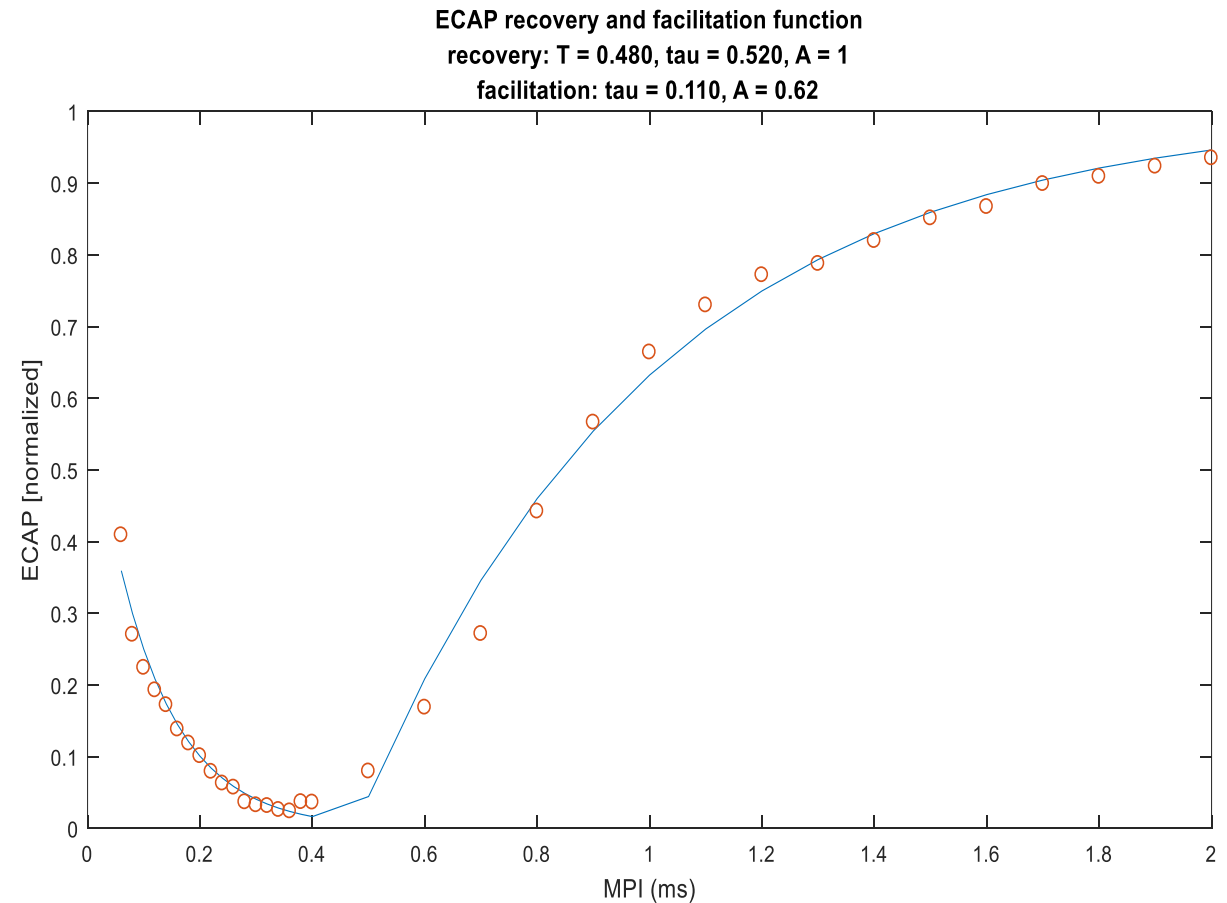


Neurons which were close to firing threshold after the 1st pulse
→ a quick enough 2nd pulse causes firing

Neurons which crossed the firing threshold after the 1st pulse
→ if 2nd pulse is (too) quick, neurons are still in refractory state

ECAP: recovery function, facilitation function

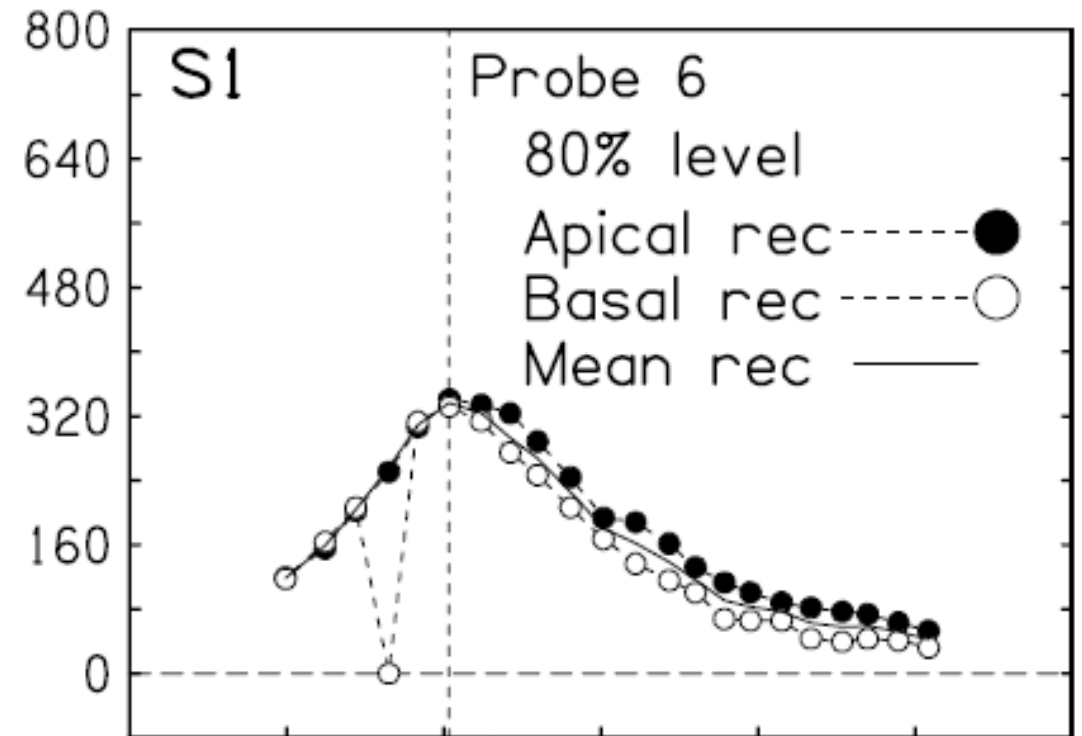
- By applying a suprathreshold masker and then a probe, the **remaining excitability** of neurons can be measured.
- **Caveat:** there are neurons further away from the electrode which were close to firing during masker pulse – if MPI is small enough, then the probe pulse will make them spike.
- Hence, facilitation function is measured at the same time!
- ARP/RRP parameters and (mostly passive) facilitation parameters obtainable!





ECAP: spread of excitation

- **NOT** measured by applying a stimulus and measuring the response at various locations!
- Reason: this measures the spread of electrical field generated by APs, not the stimulus.
- Measured by applying a masker at various locations, and a probe at a fixed location.
- This gives information on the remaining excitability at the electrode – and assuming uniform neural survival, on the overlap of masker and probe excitation fields.



Adapted from Cohen L.T. et al., Spatial spread of neural excitation in cochlear implant recipients: comparison of improved ECAP method and psychophysical forward masking, *Hear Res.* 2003 May;179(1-2):72-87.



Conclusion

- Dynamic threshold model with refractory, facilitation, accommodation and spike-rate adaptation effects implemented
- Model extended to population of neurons, included spread of excitation, refractory stochastics, developed GUI with editable neuronal and stimulus parameters, automated „synthetic“ ECAP measurements.
- **Advantages:** Phenomena in the model are able to predict responses for various modulated pulse trains: oscillations in response for square, reduction of width for sinusoidal and drop/rise in slope for saw modulated pulse trains.
- **Key deficiency:** lack of latency spread in ECAP response calculation, producing lower modulation depths than expected.
- **Remaining challenges:** resolving the above deficiency, including variable neuronal survival and thresholds, comparison and parameter fitting against bigger data set, modelling cathodic pulses.